Ayyanar Anand G

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering student (B.E., graduating 2027, CGPA 8.45) with hands-on experience in embedded systems, IoT, and Python development. Built ESP32-based multi-sensor monitoring systems, an Electric Vehicle (EV) Human Machine Interface (HMI) on Raspberry Pi using the Slint GUI framework, and AI speech and translation applications using Whisper and FFmpeg. Strong foundation in C and Python, with solid debugging, problem-solving, and collaborative development skills. Best Paper Award winner with a published patent application (No. 202641103368 A).

TECHNICAL SKILLS

Programming Languages: C, Python, SQL, Web Development, Vibe Coding, API Key Integration
Embedded Systems and IoT: ESP32, Raspberry Pi, Microcontroller and Microprocessor, IoT Development, Sensor Integration (PIR, Ultrasonic, Thermal, Sound, Vibration, ECG, MPU6050 Motion, MLX90614 Temperature), Multi-Sensor Fusion
AI and Speech Processing: Speech Recognition, Speech-to-Text, Text-to-Speech, Whisper AI, Edge TTS, Google Translate, FFmpeg
Software Engineering: Object-Oriented Programming, Debugging, GitHub, Version Control
Tools: Arduino IDE, VS Code, Tinkercad, CAD Tool
Coursework: Internet of Things, Cloud Computing, Data Analytics with Python, Industry 4.0 and Industrial IoT

EXPERIENCE

Embedded Systems Intern (2-Wheeler/3-Wheeler HMI Development) **June 2026**
Rgreen Technologies, Madurai

- Developed a Human Machine Interface (HMI) for Electric Vehicle (EV) applications on Raspberry Pi using the Slint GUI framework.
- Completed an embedded systems internship with hands-on Raspberry Pi development experience.

Research and Development Intern **January 2024 – February 2026**
Innovation Research Park (IRP) Cell, KLN College of Engineering, Madurai

- Contributed to software and IoT-based research projects involving automation and intelligent systems.
- Developed Python-based applications and collaborated with multidisciplinary engineering teams.
- Participated in technical workshops on embedded systems and product development.

PROJECTS

Smart Bird and Animal Crop Protection System **January 2026 – March 2026**
ESP32, IoT, Multi-Sensor Fusion

- Designed and developed an ESP32-based IoT monitoring system using multi-sensor fusion.
- Implemented real-time intrusion detection using PIR, ultrasonic, thermal, sound, and vibration sensors.
- Developed decision-making logic to reduce false alarms and improve detection reliability.
- **Presented the project at the INNOVATE 2026 International Conference and received the Best Paper Award.**

Smart Drowsiness Detection and Worker Behaviour Monitoring System **January 2025 – March 2026**
ESP32, ECG, MPU6050, MLX90614, Wearable Safety

- Developed a wearable ESP32-based safety prototype integrating ECG, MPU6050 motion, and MLX90614 temperature sensors for real-time drowsiness and abnormal-behaviour detection.
- Implemented buzzer and vibration-motor alerts and machine shutdown logic for industrial safety applications.
- Received ₹8,500 funding under IPC SPFS for the project.
- **Published patent application No. 202641103368 A (filed August 27, 2026; published September 4, 2026).**

VideoLingua – AI Video Voice Translator **January 2025 – February 2026**
Python, Whisper, Google Translate, Edge TTS, FFmpeg

- Developed a multilingual video translation platform using Python, Whisper, Google Translate, Edge TTS, and FFmpeg.
- Implemented speech-to-text, language translation, voice synthesis, and audio-video synchronization pipelines.
- Improved user experience through automated multilingual content generation.
- **Showcased in Sourashtra Chamber of Commerce, Madurai**

THARA – AI Voice Assistant **June 2024 – November 2024**

Python, Speech Recognition, Text-to-Speech

- Developed a Python-based voice assistant supporting speech recognition and natural language interaction.
- Implemented multilingual support (Tamil and English), text-to-speech, and information retrieval features.
- Integrated conversational AI capabilities and task automation modules.

ZYNTRIX AI – Indian Stock Market Intelligence Platform **In Progress**

AI, NVIDIA API (planned), NSE and BSE Market Data

- Building an AI-powered platform for research and analysis of NSE and BSE-listed companies, with company search and evidence-based AI explanations.
- Designing beginner-friendly explanations in Tamil and Tanglish, backend-only API key management, verified and licensed data sources, and clear labelling of demo data.

EDUCATION

B.E Electronics and Communication Engineering **September 2023 – Present (Expected 2027)**

KLN College of Engineering, Pottapalayam, Sivagangai, Tamil Nadu | CGPA: 8.45

Higher Secondary School **June 2020 – May 2022**

Vimal Matriculation and Higher Secondary School, Kalugumalai, Tuticorin, Tamil Nadu | Percentage: 78.2 %

CERTIFICATIONS

- Python Essentials 1 – Cisco Networking Academy
- Internet of Things – NPTEL
- Cloud Computing – NPTEL
- Data Analytics with Python – NPTEL
- Industry 4.0 and Industrial Internet of Things – NPTEL
- Young Professional Program (Industry Readiness and Soft Skills) – TCS iON

ACHIEVEMENTS

- Best Paper Award (₹1,000 prize) – INNOVATE 2026 International Conference, for the Smart Bird and Animal Crop Protection System
- Published Patent Application No. 202641103368 A – Smart Drowsiness Detection and Worker Behaviour Monitoring System (filed August 27, 2026; published September 4, 2026)
- Participant – Smart India Hackathon (Pre-SIH 2025), Nimirndhu Nil, Niral Thiruvizha, Seva Challenge 2026, MSME
- First Prize (₹1,500 prize) In Mini Project Competition
- First Prize (₹1,500 prize) in TECHGENIO for Idea Pitching and Hardware Expo

LANGUAGES

English, Tamil